exp262: does contacts-v1 need RoPE?

Issue [#262](https://github.com/Open-Athena/MarinFold/issues/262). contacts-v1 documents are self-describing and order-randomized — every statement carries its own `<pXXX>` coordinate and statement order is uniform noise — so the proposal is to delete RoPE and replace it with a width-3 causal *token smear*: mix the previous two tokens' embeddings into the current one, as in the nanogpt speedrun's smear module.

Phase 0 asks the trained default checkpoint whether there is anything to win, before any training is bought. Phase 1 is a 3-arm ablation at the exp232 1.5B architecture on a reduced token budget.

Two is the tight window

Enumerating the contacts-v1 grammar: a token's (statement form, slot) role is NOT a function of the previous 1 token class — `(POS, POS)` is ambiguous between a contact's second argument and a residue statement's head — and IS a deterministic function of the previous 2. Lookback 3 adds nothing. Width-3 smear is the tight bound.

What Phase 0 found

Three probes on `contacts-v1-exp199-cooldown-1.5B`, teacher-forced over ground-truth documents from the exp245 monomers, on a local A5000.

The smear half is directly motivated. Layer 1 holds two nearly pure previous-token heads (0.999 and 0.996 of their attention at offset 1) and a third splitting 0.75/0.16 across offsets 1 and 2 — a width-3 smear implemented in attention, at the cost of three heads.

The NoPE half loses its mechanism but keeps its premise. Co-referent retrieval is already distance-uniform out to 2048 tokens, so RoPE is not costing us long-range reach and the long-protein story is dead. But randomizing every exact cross-statement distance costs *less* than deterministically rescaling them at matched stretch, so the model genuinely is not reading those distances.

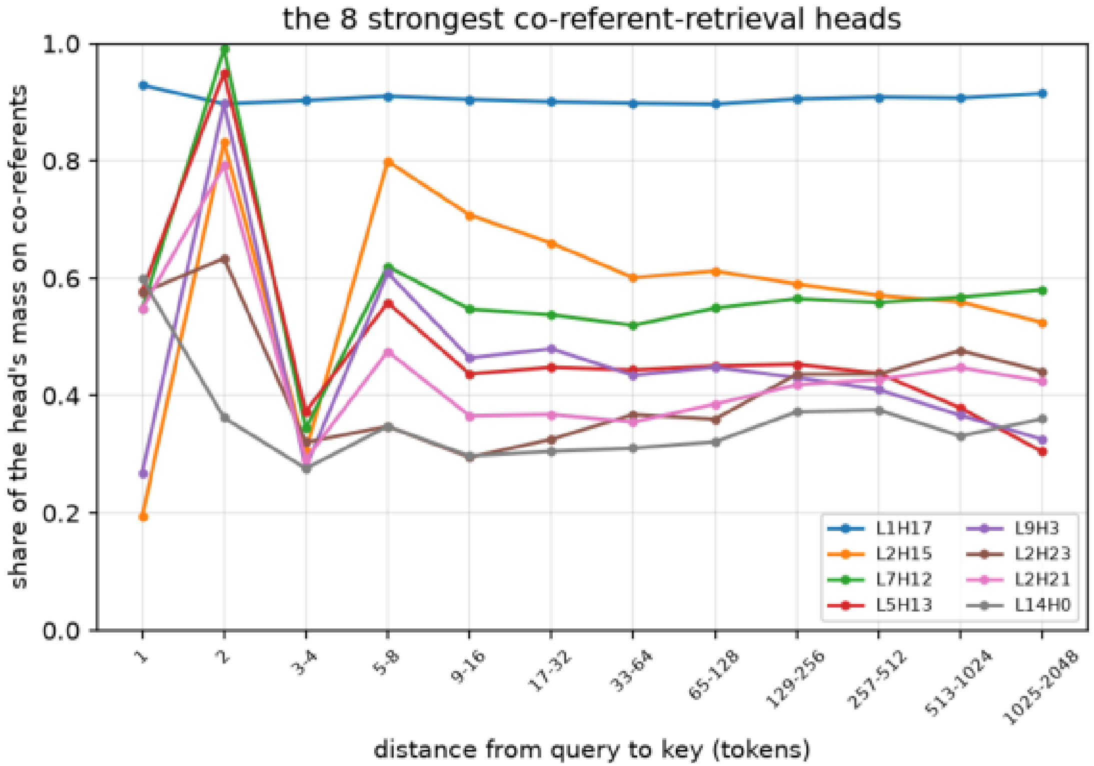
What position is actually for: counting. Keeping the position range fixed but letting two tokens share an id costs +1.23 nats, ten times any distance-randomizing intervention. The model uses position as an index, which is exactly what NoPE would take away — and exactly what the pre-registered stopping-behaviour guardrail was written to catch.

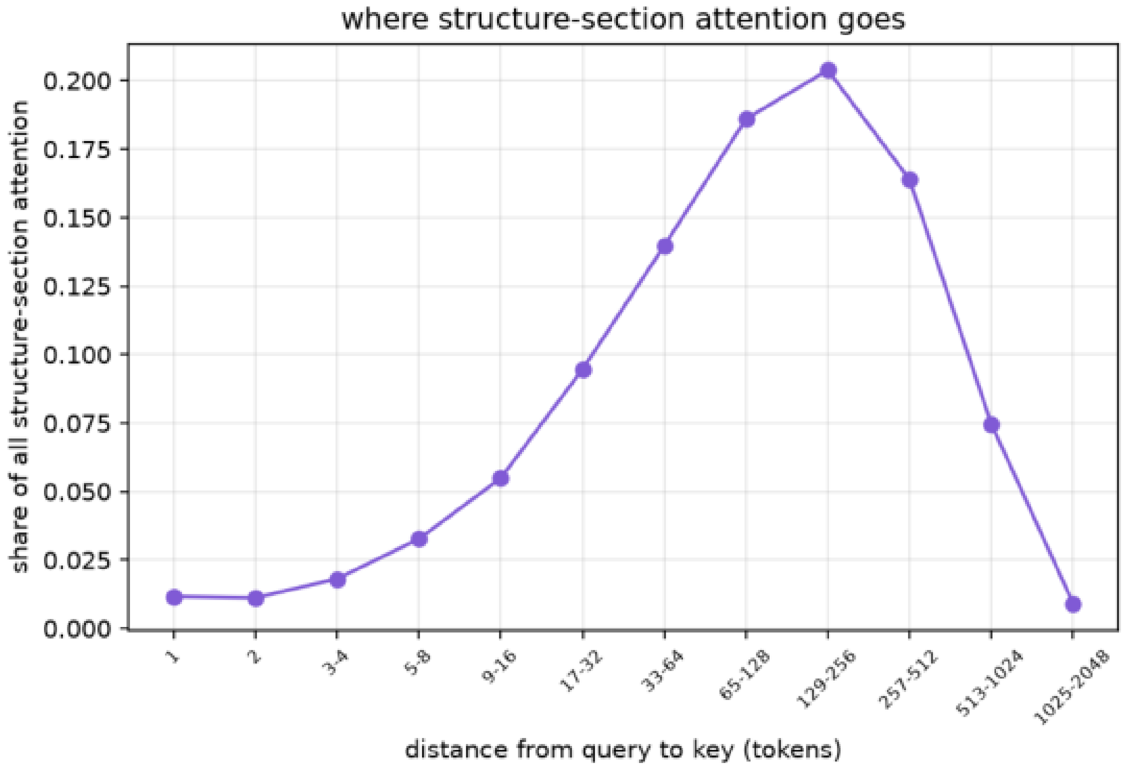
Where it stands

Phase 0 gate passed. Arm B (RoPE + smear) is promoted from safe hedge to the arm the evidence points at; arm C (NoPE + smear) proceeds with its risk localized; arm D (interleaved NoPE) is deferred as the most invasive build hedging the weakest hypothesis.

phase0_coreferent_retrieval.png

Left: for heads that retrieve earlier mentions of the query's own residue index, the share of their attention on those co-referents, by distance. Flat — L1H17 holds 0.90-0.93 out to 2048 tokens. Retrieval is already distance-uniform, so RoPE costs us no reach and the mechanistic case for dropping it fails. Right: where structure-section attention goes.

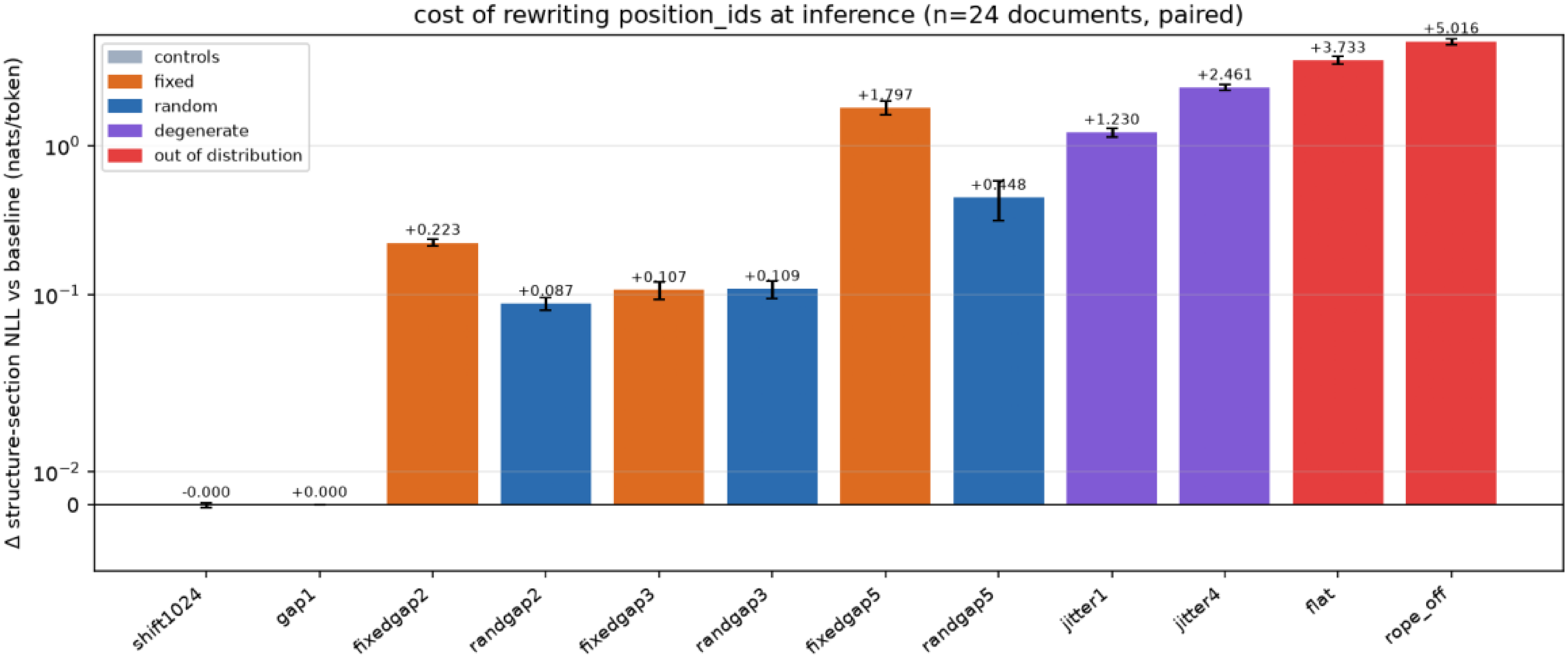




$ plot_phase0.py

phase0_position_interventions.png

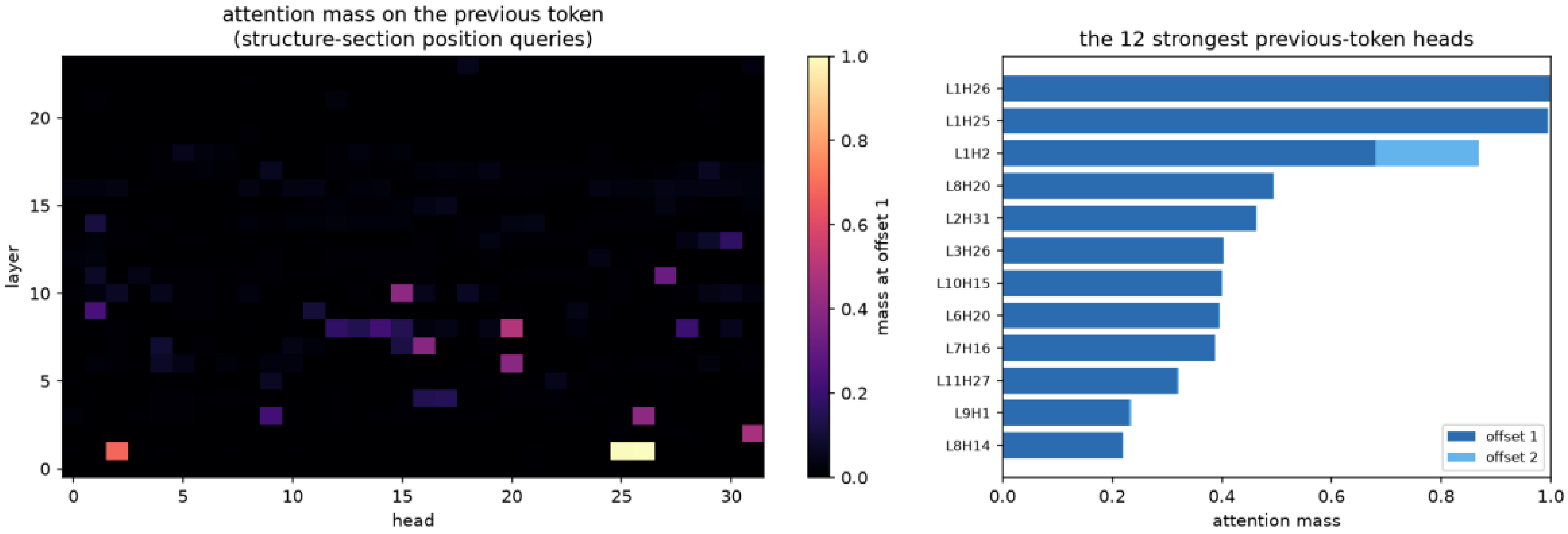
Paired change in structure-section NLL when position_ids are rewritten (n=24). fixedgapN and randgapN match on expected stretch and never repeat a position; only randgapN destroys exact cross-statement distances, and is never the worse of the pair — the model does not read them. jitterN instead repeats ids at fixed range and costs 10x more: position is an index.



$ plot_phase0.py

phase0_previous_token_heads.png

Attention mass at offset 1 per (layer, head), over structure-section position queries of 24 ground-truth documents. Layer 1 holds two near-pure previous-token heads (0.999, 0.996) and a third splitting 0.75/0.16 over offsets 1 and 2 — a width-3 smear built out of attention. Direct motivation for the smear half of #262.



$ plot_phase0.py